

Code Sample: Ohio vs. Michigan Accidental Opioid Overdose Diff-in-Diff

1 Background

The purpose of this analysis is to estimate the effect of 7-day opioid prescription limits announced in March 2017 and effective as of September 2017 on opioid death rates in Ohio. Since a randomized controlled trial is not feasible, to determine the efficacy of the treatment, I employ a differences-in-differences (DiD) approach to control for time-invariant and individual-fixed factors that would otherwise bias estimates of treatment effects. I model a simple two-group, two-period differences-in-differences estimation, where Ohio is my treatment group, and the neighboring state of Michigan is the control group. I consider August 31st, 2017 to be the date of treatment. The period of interest is November 2016 to June 2018, ten months before and after treatment, as Michigan's own 7-day prescription limits for the treatment of acute pain took effect in July 2018.

Throughout this investigation, I will describe my data sources, clean and prepare the data for analysis, conduct summary analyses of death rates in both the treatment and control groups, evaluate key assumptions for valid DiD analyses, fit a DiD model, and discuss the results as well as any limitations of the analysis.

2 Analysis

2.1 Data

I use the CDC WONDER Multiple Cause of Death data, restricted to substance related underlying causes, to compare track opioid death rates in Ohio and Michigan in the ten months before and after September 2017.

2.1.1 Importing and Merging Data

Due to export size restrictions, I exported the data in 2-4 year subsets. Hence, I import multiple files and merge them below.

```
cod_raw <- list.files("/projects/[REDACTED]/import/[REDACTED]/",
  , pattern = "Multiple Cause of Death"
  , full.names = TRUE) %>%
  map_dfr(read_csv) %>% # import and combine files in folder
  clean_names() # clean variable names

# view sample of the data and variables available
head(cod_raw)

## # A tibble: 6 x 15
##   notes state state_code year year_code month month_code multiple_cause_of_de-1
##   <chr> <chr> <chr>      <dbl>      <dbl> <chr> <chr>      <chr>
## 1 <NA> Alab~ 01        2012        2012 Jan.~ 2012/01   Other and unspecified-
## 2 <NA> Alab~ 01        2012        2012 Feb.~ 2012/02   Other and unspecified-
## 3 <NA> Alab~ 01        2012        2012 Mar.~ 2012/03   Other and unspecified-
## 4 <NA> Alab~ 01        2012        2012 Apr.~ 2012/04   Other and unspecified-
## 5 <NA> Alab~ 01        2012        2012 May,~ 2012/05   Other and unspecified-
## 6 <NA> Alab~ 01        2012        2012 Jun.~ 2012/06   Other and unspecified-
## # i abbreviated name: 1: multiple_cause_of_death
```

```
## # i 7 more variables: multiple_cause_of_death_code <chr>,
## #   underlying_cause_of_death <chr>, underlying_cause_of_death_code <chr>,
## #   deaths <dbl>, population <chr>, crude_rate <chr>, age_adjusted_rate <chr>
```

From this data, I have variables indicating state, year, month, multiple cause of death, underlying cause of death, and number of deaths.

Values appear to be unavailable at the level of the data for population, crude death rate, and age-adjusted death rate.

2.1.2 Investigating Data Quality and Availability

First, I confirm that there are no duplicate observations at the state code, month code, multiple cause of death code, and underlying cause of death code level, which is a risk I incur when exporting data subsets and re-merging.

```
# check for duplicates

check_duplicates <- cod_raw %>%
  select(contains("code")) %>%
  group_by_all() %>%
  summarize(ct = n()) %>%
  filter(ct > 1)

print(check_duplicates)
```

```
## # A tibble: 1 x 6
## # Groups:   state_code, year_code, month_code, multiple_cause_of_death_code [1]
##   state_code year_code month_code multiple_cause_of_dea~1 underlying_cause_of_~2
##   <chr>         <dbl> <chr>      <chr>                <chr>
## 1 <NA>          NA <NA>      <NA>                <NA>
## # i abbreviated names: 1: multiple_cause_of_death_code,
## #   2: underlying_cause_of_death_code
## # i 1 more variable: ct <int>
```

Duplicate observations appear to occur only when the values for all variables are null. I plan to filter these out in the cleaning step.

Next, I investigate the states and years the data cover.

```
distinct_years <- cod_raw %>%
  select(year) %>%
  unique() %>%
  print()
```

```
## # A tibble: 12 x 1
##   year
##   <dbl>
## 1 2012
## 2  NA
## 3 2013
## 4 2014
## 5 2015
## 6 2016
## 7 2017
## 8 2018
## 9 2019
## 10 2020
```

```
## 11 2010
## 12 2011

distinct_states <- cod_raw %>%
  select(state) %>%
  unique() %>%
  print()
```

```
## # A tibble: 49 x 1
##   state
##   <chr>
## 1 Alabama
## 2 Arizona
## 3 Arkansas
## 4 California
## 5 Colorado
## 6 Connecticut
## 7 Florida
## 8 Georgia
## 9 Idaho
## 10 Illinois
## # i 39 more rows
```

The data cover 48 states (excluding NA), including Ohio and Michigan, and 11 years (excluding NA) from 2012 to 2020.

Then, I identify how many missing values each variable has.

```
missing_cts <- colSums(is.na(cod_raw)) %>%
  data.frame() %>%
  filter(. > 0) %>%
  print()
```

```
##
## notes                31263
## state                542
## state_code           542
## year                 542
## year_code            542
## month                542
## month_code           542
## multiple_cause_of_death 542
## multiple_cause_of_death_code 542
## underlying_cause_of_death 542
## underlying_cause_of_death_code 542
## deaths              542
## population           542
## crude_rate           542
## age_adjusted_rate     542
```

As expected from the duplicate investigation, each variable has 542 missing values, except for *notes*, which has 31,263 missing values and will not be used in the analysis anyways.

2.2 Clean and Prepare Data for Analysis

In this next part, I clean and prepare my data for analysis. This includes 1. filtering the data to the states, periods, and death causes of interest, 2. aggregating deaths across underlying causes, 3. transforming variables

as needed, 4. joining ALFRED state-level population data to calculate death rates, and 5. creating time and treatment dummies to be used in the diff-in-diff analyses.

Below, I filter my data to Ohio and Michigan in the periods spanning December 2016 to June 2018. In addition, I limit underlying causes of death to ICD-10 underlying cause of death codes X40-X44 and multiple causes of death to opioid-related ICD multiple cause of death codes: T40.1, T40.2, T40.3, T40.4, and T40.6, as is standard* in analyses of accidental opioid overdoses.

*<https://ibis.utah.gov/ibisph-view/pdf/resource/query/DrugOverdoseMortalityCodes.pdf>

```
cod_clean <- cod_raw %>%
  mutate(month = as.Date(paste0(year, substring(month_code, 6, 7), "01"), format="%Y%m%d")) %>%
  filter(
    # filter periods
    month >= "2016-11-01"
    , month <= "2018-06-01"
    # filter groups
    , state %in% c("Ohio", "Michigan")
    # limit to accidental poisoning & opioid-related death codes
    , str_detect(underlying_cause_of_death_code, "^X4[0-4]$")
    , multiple_cause_of_death_code
      %in% c("T40.1", "T40.2", "T40.3", "T40.4", "T40.6")
  ) %>%
  mutate(
    state
    , year
    , month
    , multiple_cause_of_death
    , multiple_cause_of_death_code
    , deaths) %>%
  group_by(state, year, month
    , multiple_cause_of_death
    , multiple_cause_of_death_code) %>%
  # aggregate opioid deaths
  summarize(deaths = sum(deaths)) %>%
  ungroup()
```

Note that in the above chunk, I aggregated opioid deaths across underlying causes of death. This does not introduce potential for multi-counting certain deaths, since each death certificate lists a single underlying cause of death, and, thus, the groups are mutually exclusive. However, if I were to aggregate across multiple causes of death, I would run the risk of double counting deaths, for more than one multiple cause of death can be associated with more than one multiple causes of death. In fact, the risk is even high considering the context of the analysis being drug overdoses, where more than one drug is often found in a person's system in the event of an overdose. See <https://wonder.cdc.gov/wonder/help/ucd.html> for more information on how ICD-10 codes map to death certificates.

Furthermore, I inspect the availability of ICD-10 code reporting in Ohio and Michigan during the analysis period.

```
check_cod_availability <- cod_clean %>%
  filter(state %in% c("Ohio", "Michigan")) %>%
  group_by(state, multiple_cause_of_death) %>%
  summarize(ct = n()) %>%
  ungroup() %>%
  print()
```

`summarise()` has grouped output by 'state'. You can override using the

```
## `.groups` argument.
## # A tibble: 8 x 3
##   state      multiple_cause_of_death      ct
##   <chr>      <chr>                    <int>
## 1 Michigan Heroin                      20
## 2 Michigan Methadone                   3
## 3 Michigan Other opioids                20
## 4 Michigan Other synthetic narcotics   20
## 5 Ohio      Heroin                      20
## 6 Ohio      Other and unspecified narcotics 2
## 7 Ohio      Other opioids                20
## 8 Ohio      Other synthetic narcotics   20
```

Reporting on Methadone (T40.3) and Other Unspecified Narcotics (T40.6) is limited during the period of analysis period for both states. Though, both Ohio and Michigan report death counts for Heroin, Other Opioids, and Other Synthetic Narcotics in all 20 periods. If multiple cause of death ICD-10 groups were mutually exclusive, I would consider aggregating these groups, however, as previously discussed, we can not aggregate without inviting double counting risks. Yet more, aggregation poses conceptual risks, as there are meaningful differences between these groups that may obfuscate from the true treatment effects at a given time. For example, Heroin use often occurs later in the cycle of opioid addiction, so changes in Heroin-related accidental drug deaths may take longer to appear as a result of an intervention at an initial prescription. Also, by forfeiting this variation, I would lose the ability to analyze the role of substitution as a continuation of this analysis (i.e. Do prescription caps lead those experiencing addiction to substitute Heroin for Other Opioids?).

Thus, I elect to keep the codes disaggregated, but I restrict my analysis to the three ICD-10 groups that are completely represented across all periods.

Now, I join the data to the ALFRED population data. Since the ALFRED population data is only available by state at the annual-level, I assume linear population growth to interpolate the state populations in a given month. Note that the ALFRED population data is referenced to July 1.

```
alfred <- read_xlsx("/projects/[REDACTED]/import/[REDACTED]/US_population_by_state.xlsx"
                  , sheet = "Annual") %>%
  # add buffer years for lead and lag calculations
  filter(year(observation_date) >= 2015
         , year(observation_date) <= 2020) %>%
  pivot_longer(cols = contains("POP")
               , names_to = "state"
               , values_to = "population") %>%
  transmute(year = year(observation_date)
           , state = state.name[match(str_remove(state, "POP"), state.abb)]
           , population = population*1000) %>%
  arrange(year) %>%
  group_by(state) %>%
  mutate(population_lead = lead(population, 1)
         , population_lag = lag(population, 1))

cod_joined <- cod_clean %>%
  filter(multiple_cause_of_death_code
        %in% c("T40.1", "T40.2", "T40.4")) %>%
  left_join(alfred, by = c("year", "state")) %>%
  # transform population in a month as a weighted average of
  # the distance between two annual population estimates
  mutate(
```

```

# subtract 7 from month to anchor July as 0
month_num = as.numeric(month(month)) - 7
# no change to July population
, population = case_when(month_num == 0 ~ population
  # for August-December, weight by distance from
  # population estimate in current and next year
  , month_num > 0 ~ (month_num*population_lead
    + (12-month_num)*population)/12
  # for January-June, weight by distance from
  # population estimate in current and prior year
  , month_num < 0 ~ (-month_num*population_lag
    + (12+month_num)*population)/12)
, death_rate_100k = 100000*deaths/population) %>%
select(state
  , month
  , multiple_cause_of_death
  , deaths
  , population
  , death_rate_100k)

```

Then, I add my treatment and control dummies to prepare the data for the DiD analyses.

```

cod_diff_in_diff <- cod_joined %>%
  mutate(time_dum = if_else(month >= "2017-09-01", 1, 0)
    , treated_dum = if_else(state == "Ohio", 1, 0))

```

3 Summary Analyses

3.1 Summary Table of Death Rates

Before performing regression analyses, I pull summary statistics of death rates by state, pre- vs. post-treatment, and ICD-10 multiple cause of death codes.

```

death_rate_summary_stats <- cod_diff_in_diff %>%
  group_by(multiple_cause_of_death, time_dum, state) %>%
  summarize(n = n()
    , mean = mean(death_rate_100k, na.rm=TRUE)
    , sd = sqrt(mean((death_rate_100k
      - mean(death_rate_100k, na.rm=TRUE))^2
      , na.rm=TRUE))
    , median = median(death_rate_100k, na.rm=TRUE)
    , min = min(death_rate_100k, na.rm = TRUE)
    , max = max(death_rate_100k, na.rm = TRUE)) %>%
  ungroup() %>%
  print()

```

`summarise()` has grouped output by 'multiple_cause_of_death', 'time_dum'. You
can override using the `.groups` argument.

```

## # A tibble: 12 x 9
##   multiple_cause_of_death time_dum state      n mean    sd median  min  max
##   <chr>                  <dbl> <chr> <int> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Heroin                0 Mich~    10 0.647 0.0814 0.652 0.521 0.762
## 2 Heroin                0 Ohio     10 0.800 0.169 0.776 0.531 1.06
## 3 Heroin                1 Mich~    10 0.541 0.0804 0.536 0.411 0.671

```

## 4 Heroin	1 Ohio	10	0.546	0.0957	0.557	0.394	0.694
## 5 Other opioids	0 Mich~	10	0.385	0.0987	0.371	0.261	0.592
## 6 Other opioids	0 Ohio	10	0.679	0.168	0.631	0.446	1.00
## 7 Other opioids	1 Mich~	10	0.325	0.0566	0.351	0.220	0.421
## 8 Other opioids	1 Ohio	10	0.375	0.0495	0.373	0.283	0.489
## 9 Other synthetic narcoti~	0 Mich~	10	1.03	0.163	1.02	0.752	1.30
## 10 Other synthetic narcoti~	0 Ohio	10	2.46	0.441	2.36	1.77	3.02
## 11 Other synthetic narcoti~	1 Mich~	10	1.22	0.134	1.19	0.981	1.46
## 12 Other synthetic narcoti~	1 Ohio	10	1.97	0.236	1.95	1.60	2.45

For Heroin-related deaths, it appears that the mean death rate is closer in magnitude after treatment than before treatment. For Other Opioids and Other Synthetic Opioids, this is also true.

I also perform a cursory evaluation of pre-treatment summary statistics.

```
pre_trend_investigation <- cod_diff_in_diff %>%
  filter(time_dum == 0) %>%
  group_by(multiple_cause_of_death, state) %>%
  summarize(mean_death_rate = mean(death_rate_100k, na.rm=TRUE)
            , mean_population = mean(population, na.rm=TRUE)
            # calculate death rate linear trend
            , death_rate_slope = (last(death_rate_100k)
                                - first(death_rate_100k))/(n()-1)) %>%
  print()
```

```
## # A tibble: 6 x 5
## # Groups:   multiple_cause_of_death [3]
##   multiple_cause_of_death state mean_death_rate mean_population death_rate_slope
##   <chr>                <chr>      <dbl>          <dbl>          <dbl>
## 1 Heroin               Mich~      0.647          9970049.         0.00436
## 2 Heroin               Ohio      0.800         11658139.        -0.0459
## 3 Other opioids        Mich~      0.385          9970049.        -0.00566
## 4 Other opioids        Ohio      0.679         11658139.        -0.0287
## 5 Other synthetic narcot~ Mich~      1.03          9970049.         0.0165
## 6 Other synthetic narcot~ Ohio      2.46         11658139.         0.0130
```

Crucially, upon visual inspection of the numbers, the linear trends of Michigan and Ohio appear most parallel for Other Synthetic Narcotics. However, I do not perform any tests at this point to evaluate the statistical strength of this relationship.

4 Evaluate Key Assumptions

Some necessary conditions for DiD estimation to be valid include the following:

1. Pre-treatment trends are parallel.
2. The Stable Unit Treatment Value Assumption (SUTVA) holds.

I perform statistical tests to evaluate item 1 and provide a conceptual discussion of item 2.

4.1 Assumption 1: Parallel Trends

As a two-group, two-period implementation of a two-way fixed effects model, a DiD analysis requires that there are no confounding variables that vary with time and between groups. In other words, parallel trends hold. For example, if Michigan were to (and did in fact) expand access to opioid counseling in the time between September 2017 and July 2018, then the DiD method would not account for this factor and the estimated average treatment effect would be biased (omitted variable bias). In a formal research investigation,

I would research policy and demographic changes and control for any time-varying differences between the groups in my regression.

4.1.1 Trend Visualizations

To verify parallel trends, or, at the very least, not rule them out, I chart death rates in Ohio and Michigan over time, and perform a visual inspection.

```
chart_input_df <- data.frame(cause_of_death =
  c("T40.1" = "Heroin"
    , "T40.2" = "Other opioids"
    , "T40.4" = "Other synthetic narcotics"))

chart_ts <- function(cause_of_death){

  chart_data <- cod_diff_in_diff %>%
    filter(multiple_cause_of_death == cause_of_death)

  y_min = floor(min(chart_data$death_rate_100k, na.rm = TRUE))

  y_max = ceiling(max(chart_data$death_rate_100k, na.rm = TRUE))

  timeseries <-
    ggplot(chart_data
    ) +
    geom_line(aes(x = month
      , y = death_rate_100k
      , color = state
      , group = state)
    ) +
    scale_y_continuous(expand = c(0, 0)
      , breaks = seq(y_min, y_max, by = 0.2)
      , limits = c(y_min, y_max)
    ) +
    scale_x_date(expand = c(0, 0)
      , date_breaks = "1 month"
      , date_labels = "%b %Y"
    ) +
    geom_vline(xintercept = as.Date("2017-09-01")
      , linetype = "longdash"
    ) +
    annotate("text"
      , x = as.Date("2017-09-01")
      , y = Inf
      , label = "Ohio Prescription Limit Effective"
      , angle = 90
      , family = "serif"
      , vjust = -.5
      , hjust = 1
      , size = 3.5
      , color = "black"
    ) +
    scale_color_manual(values = c('Ohio' = '#E41A1C', 'Michigan' = '#254061'))
  ) +
  labs(title = glue("Opioid Related Deaths in Michigan vs. Ohio"))
```

```

    , subtitle = glue("{str_to_title(cause_of_death)}, Nov 2016 - Jun 2018")
    , y = glue("Deaths per 100k")
    , x = ""
  ) +
  theme_classic() +
  theme(text = element_text(family = "serif", size = 10)
    , plot.title = element_text(hjust = 0.5, face = "bold", size = 16)
    , plot.subtitle = element_text(hjust = 0.5, size = 12)
    , axis.text.x = element_text(angle = 60, hjust = 1, size = 8)
    , axis.text.y = element_text(size = 9)
    , axis.title.y = element_text(size = 11)
    , legend.position = "bottom"
    , legend.box = "horizontal"
    , legend.title = element_blank()
  )
}

charts <- pmap(chart_input_df, chart_ts)

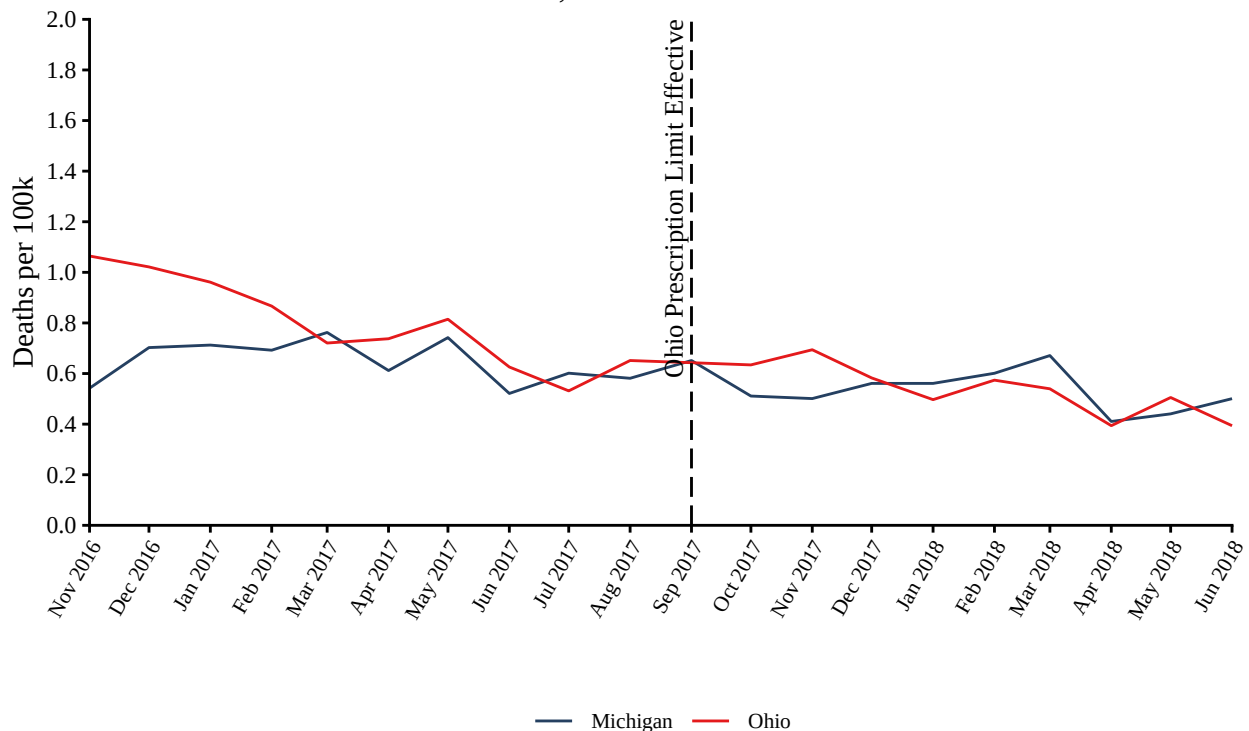
# Review trends in Heroin-related accidental overdose deaths

heroin_trends <- charts[[1]] %>%
  plot()

```

Opioid Related Deaths in Michigan vs. Ohio

Heroin, Nov 2016 – Jun 2018

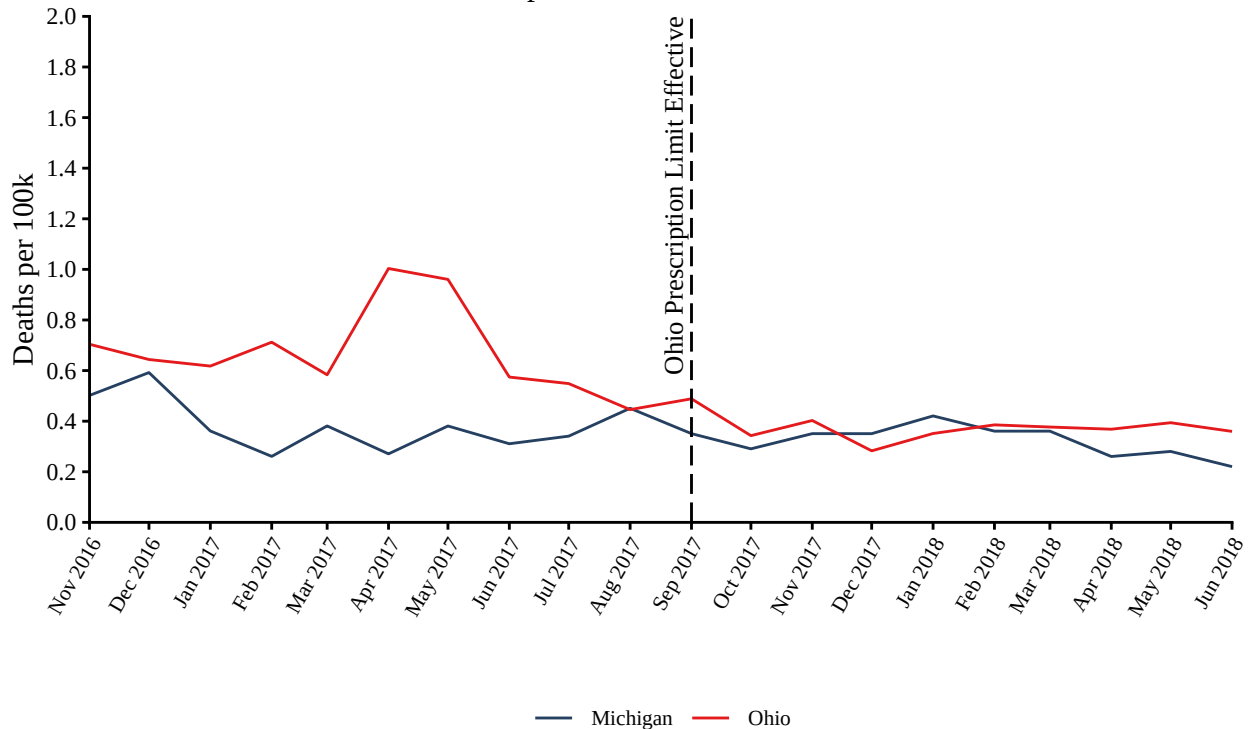


Upon visual inspection, trends appear to be generally on par in terms of Heroin deaths in the period leading up to September 2017. Notably, they seem to diverge slightly in the few months after prescription limits are effective but converge around December 2017.

```
# Review trends in Other Opioid-related accidental overdose deaths
other_opioid_trends <- charts[[2]] %>%
  plot()
```

Opioid Related Deaths in Michigan vs. Ohio

Other Opioids, Nov 2016 – Jun 2018



Trends appear somewhat parallel until January 2017 when they diverge sizeably and then dramatically after March 2017. Strikingly, death rates appear to converge and then diverge again in the period prior to treatment. Trends appear more parallel after treatment than before.

I also notice a spike in Other Opioid death rates in Ohio in April 2017 that persists for two months until June 2017. There may be some sort of external shock that is affecting Other Opioid death rates in this period. In a more expansive analysis, I would likely conduct some public research to see if I can explain and attempt to control for the factors driving the pre-treatment spike.

```
# Review trends in Other Synthetic Narcotic-related accidental overdose deaths

synthetic_trends <- charts[[3]] %>%
  plot()
```

Opioid Related Deaths in Michigan vs. Ohio

Other Synthetic Narcotics, Nov 2016 – Jun 2018



Trends in Other Synthetic Narcotic death rates appear somewhat parallel, barring a few spikes in Ohio's death rate, up until treatment and two months past treatment. After November 2017, Other Synthetic Narcotic-related death rates begin trending downward and flattening out. A delayed treatment effect may make sense as this category is primarily composed of Fentanyl deaths; Fentanyl is typically used in later stages of the course of addiction. There are several sharp spikes that indicate there may be a seasonal component to deaths or even reporting of the deaths. I considered using an MA(3) transformation of death rates to smooth the trend, but, due to concerns about limitations to inference introduced by serial correlation in a two-group setting (unable to cluster SE's), I elected to continue with raw death rates.

4.1.2 Statistical Tests of Trends

Now, I perform a simple linear regression test to see if parallel trends are violated. In particular, I test whether there is a significant interaction between the month t and group i (state) in the pre-treatment period.

The model treats deaths per 100,000 as a linear function of treatment period and treatment group (Ohio) dummies as well as the interaction between the two dummies:

$$DeathRate100k_{ti} = \beta_0 + \beta_1 * Month_t + \beta_2 * Ohio_i + \beta_3 * Month_t \times Ohio_i \quad t \in \{1, 2, \dots, Treatment\ Month - 1\}, \quad i \in \{0, 1\}$$

```
model_pre_trends <- function(cause_of_death){
  # filter to pre-treatment data
  pre_trend_data <- cod_diff_in_diff %>%
    filter(multiple_cause_of_death == cause_of_death
           , time_dum == 0) %>%
  # get time period
  mutate(time_num = match(month, unique(cod_diff_in_diff$month)) - 1)
```

```

# evaluate whether we can reject null hypothesis on the interaction
# between the period and the group:  $B = 0$ 
model <- lm(death_rate_100k ~ time_num * treated_dum, data = pre_trend_data)

summary(model)

}

pre_trends <- pmap(chart_input_df, model_pre_trends)

```

I am testing $H_0 : \beta_3 = 0$ and $H_a : \beta_3 \neq 0$. If H_0 is rejected, then parallel trends are violated. If I fail to reject H_0 , then the outcome of this test is consistent with parallel trends, but it is not definitive. In particular, failure to reject a null hypothesis does not imply and acceptance of the alternative hypothesis.

```

heroin_pre_trends <- pre_trends[[1]] %>%
  print()

```

```

##
## Call:
## lm(formula = death_rate_100k ~ time_num * treated_dum, data = pre_trend_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.140660 -0.035529  0.003068  0.036445  0.111422
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.682738   0.046897  14.558 1.20e-10 ***
## time_num       -0.007953   0.008785  -0.905  0.37872
## treated_dum     0.361212   0.066322   5.446 5.38e-05 ***
## time_num:treated_dum -0.046364  0.012423  -3.732  0.00182 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07979 on 16 degrees of freedom
## Multiple R-squared:  0.7818, Adjusted R-squared:  0.7409
## F-statistic: 19.11 on 3 and 16 DF,  p-value: 1.539e-05

```

For Heroin, I reject H_0 . β_3 is significant at the 99% confidence level (P-Value = 0.002).

```

other_opioid_pre_trends <- pre_trends[[2]] %>%
  print()

```

```

##
## Call:
## lm(formula = death_rate_100k ~ time_num * treated_dum, data = pre_trend_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.17797 -0.08869 -0.03594  0.02608  0.33031
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.445721   0.087252   5.108 0.000105 ***
## time_num       -0.013455   0.016344  -0.823 0.422483
## treated_dum     0.289346   0.123393   2.345 0.032258 *

```

```
## time_num:treated_dum 0.001078 0.023114 0.047 0.963386
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1484 on 16 degrees of freedom
## Multiple R-squared: 0.5663, Adjusted R-squared: 0.4849
## F-statistic: 6.963 on 3 and 16 DF, p-value: 0.003276
```

For Other Opioids, I fail to reject the null hypothesis. β_3 is insignificant and close to zero.

```
synthetic_pre_trends <- pre_trends[[3]] %>%
  print()
```

```
##
## Call:
## lm(formula = death_rate_100k ~ time_num * treated_dum, data = pre_trend_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.73876 -0.14090 -0.06861  0.21616  0.55275
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.059737   0.217863   4.864 0.000172 ***
## time_num         -0.006587   0.040809  -0.161 0.873798
## treated_dum       1.447472   0.308105   4.698 0.000242 ***
## time_num:treated_dum -0.003311   0.057713  -0.057 0.954955
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3707 on 16 degrees of freedom
## Multiple R-squared: 0.8237, Adjusted R-squared: 0.7907
## F-statistic: 24.92 on 3 and 16 DF, p-value: 2.858e-06
```

For Other Synthetic Narcotics, I fail to reject the null hypothesis. β_3 is insignificant and close to zero.

Furthermore, I considered an event study approach to pre-testing (He & Wang, 2017). Though, it is possible that there are alternative functional forms that would output similar pre-trend coefficients but yield substantial bias (Roth, 2019).

While I can neither verify or reject parallel trends outright, I continue with this assumption for Other Opioids and Other Synthetic Narcotics for the sake of the exercise. I recognize the limitations this imposes on the validity of the results.

4.2 Assumption 2: SUTVA

An unbiased DiD analysis would require that the composition of the comparison groups remain stable over the analysis period. In addition, the treatment of one group should not affect outcomes in the other group. In this case, we must assume that people did not move in and out of Michigan or Ohio in any way that would impact opioid-related death rates. One other essential consideration in analyses of addiction or any sort of epidemiological question, for that matter, is whether or not the remaining living population is equally susceptible to addiction. In other words, if the majority of people who are prone to opioid addiction or vulnerable to poor prescription practices in Ohio have already died of opioid overdoses, then we may see death rates decrease, even though it would not be a result of the policy intervention. Moreover, spillover effects are possible, since Michigan and Ohio are neighboring states. Those suffering from an opioid addiction in Ohio may cross state lines to seek prescriptions if they are denied their desired quantity. Or, if fewer

people are addicted to opioids in Ohio, producers and sellers of Heroin and Fentanyl may turn to larger, nearby markets, like Michigan to maintain or increase sales.

Despite the limitations discussed, I proceed with the analysis taking these assumptions as given for the sake of the exercise.

5 Fitting the Differences-in-Differences Model

For illustrative purposes, I derive the average treatment effect (ATE) of opioid prescription limits using an algebraic and a regression method. Both methods should yield the same results.

5.1 Algebraic ATE Estimation

Conceptually, the average treatment effect in a DiD model is the difference between (between estimator) the difference between the pre- and post- treatment means within each group (within estimator).

The ATE is algebraically defined as follows:

$$\text{Average Treatment Effect} = (Y_{Ohio, 1} - Y_{Ohio, 0}) - (Y_{Michigan, 1} - Y_{Michigan, 0}), \quad 1 = \text{On or after September 2017}$$

```
get_ate <- function(cause_of_death){

yt1 = cod_diff_in_diff %>%
  filter(treated_dum == 1, time_dum == 1
         , multiple_cause_of_death == cause_of_death) %>%
  pull(death_rate_100k) %>%
  mean(., na.rm = TRUE)

yt0 = cod_diff_in_diff %>%
  filter(treated_dum == 1, time_dum == 0
         , multiple_cause_of_death == cause_of_death) %>%
  pull(death_rate_100k) %>%
  mean(., na.rm = TRUE)

yc1 = cod_diff_in_diff %>%
  filter(treated_dum == 0, time_dum == 1
         , multiple_cause_of_death == cause_of_death) %>%
  pull(death_rate_100k) %>%
  mean(., na.rm = TRUE)

yc0 = cod_diff_in_diff %>%
  filter(treated_dum == 0, time_dum == 0
         , multiple_cause_of_death == cause_of_death) %>%
  pull(death_rate_100k) %>%
  mean(., na.rm = TRUE)

treatment_effect = (yt1 - yt0) - (yc1 - yc0)

}

input_df <- data.frame(cause_of_death =
  c("T40.2" = "Other opioids"
    , "T40.4" = "Other synthetic narcotics"))
```

```
ate <- pmap(input_df, get_ate)
```

```
ate
```

```
## [[1]]
## [1] -0.2435739
##
## [[2]]
## [1] -0.6766033
```

The estimated ATE of the treatment on Other Opioid-related accidental overdose deaths is -0.24. For Other Synthetic Narcotics, the ATE is -0.68. I interpret and discuss these results in the next section.

5.2 Regression ATE Estimation

Now, I fit a regression model to estimate the ATE of the DiD.

Death Rates $100k_{ij} = \beta_0 + \beta_1 * State_i + \beta_2 * Treatment\ Period_j + \beta_3 * State_i \times Treatment\ Period_j + \varepsilon_{ij}$, $i, j \in \{0, 1\}$

```
get_est <- function(cause_of_death){
  model_data <- cod_diff_in_diff %>%
    filter(multiple_cause_of_death == cause_of_death)

  model <- lm(death_rate_100k ~ time_dum + treated_dum + time_dum:treated_dum
    , data = model_data)

  summary(model)
}

diff_in_diff <- pmap(input_df, get_est)

lm_other_opioids = diff_in_diff[[1]] %>%
  print()

##
## Call:
## lm(formula = death_rate_100k ~ time_dum + treated_dum + time_dum:treated_dum,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23367 -0.06217 -0.00555  0.02883  0.32412
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.38518    0.03483   11.059 3.94e-13 ***
## time_dum       -0.06060    0.04925   -1.230  0.22656
## treated_dum     0.29420    0.04925    5.973 7.55e-07 ***
## time_dum:treated_dum -0.24357    0.06966   -3.497  0.00127 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.1101 on 36 degrees of freedom
## Multiple R-squared:  0.6406, Adjusted R-squared:  0.6106
## F-statistic: 21.39 on 3 and 36 DF,  p-value: 3.979e-08
```

Like the algebraic representation, $\widehat{ATE} = \hat{\beta}_3 = -0.24$, for Other Opioid-related accidental overdose deaths. This result is significant at the 99% Confidence Level (P-Value = 0.0013).

Given all necessary DiD assumptions hold, we could conclude that Ohio's 7-Day opioid prescription limit for acute pain had the effect of reducing accidental Other Opioid-related overdose deaths by 0.24 deaths per 100k.

```
average_post_treatment_pop <- cod_joined %>%
  filter(state == "Ohio"
    , month >= "2017-09-01") %>%
  summarize(mean = mean(population, na.rm = TRUE)) %>%
  pull()

crude_other_opioid <- (ate[[1]]/100000)*average_post_treatment_pop

crude_other_opioid
```

```
## [1] -28.43465
```

Adjusted for Ohio's average population from September 2017 to June 2018, this translates to an average of 28.4 fewer Other Opioid-related deaths per month.

```
ate_lm_synthetic_narcotics = diff_in_diff[[2]] %>%
  print()
```

```
##
## Call:
## lm(formula = death_rate_100k ~ time_dum + treated_dum + time_dum:treated_dum,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.69422 -0.13364 -0.03869  0.16514  0.55804
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.03010    0.09051   11.382 1.75e-13 ***
## time_dum          0.18504    0.12799    1.446 0.156911
## treated_dum       1.43257    0.12799   11.193 2.82e-13 ***
## time_dum:treated_dum -0.67660    0.18101   -3.738 0.000643 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2862 on 36 degrees of freedom
## Multiple R-squared:  0.8191, Adjusted R-squared:  0.804
## F-statistic: 54.34 on 3 and 36 DF,  p-value: 1.911e-13
```

Like the algebraic representation, $\widehat{ATE} = \hat{\beta}_3 = -0.68$, for Other Synthetic Narcotic-related accidental overdose deaths. This result is significant at the 99.9% Confidence Level (P-Value = 0.0006).

Given all necessary DiD assumptions hold, we could conclude that Ohio's 7-Day opioid prescription limit for acute pain had the effect of reducing accidental Other Synthetic Narcotic-related overdose deaths by 0.68 deaths per 100k.

```
crude_synthetic_narcotics <- (ate[[2]]/100000)*average_post_treatment_pop  
crude_synthetic_narcotics
```

```
## [1] -78.98621
```

Adjusted for Ohio's average population from September 2017 to June 2018, this translates to an average of 79 fewer Other Synthetic Narcotic-related deaths per month.

Both of these results being negative is directionally consistent with what we would expect from an intervention intended to decrease opioid deaths.

6 Limitations & Discussion

I conclude that, given all assumptions hold for the sake of the exercise, the causal effects of Ohio's policy intervention on Other Opioid- and Other Synthetic Narcotic-related death rates were negative and significant. One additional limitation, which I have not yet discussed, is the fact that Ohio's new prescription limit policy was announced in March 2017, six months in advance of its effective date. This means that, in the time leading up to treatment, while not required to do so, clinicians may have begun complying with these guidelines to align with professional standards. As a result, the effects of this policy may have spilled over into these earlier months. This could invalidate the conclusions I drew from the parallel trends analyses and mean that my ATE estimates I reported are biased in such a way that they are smaller in magnitude than the true values.